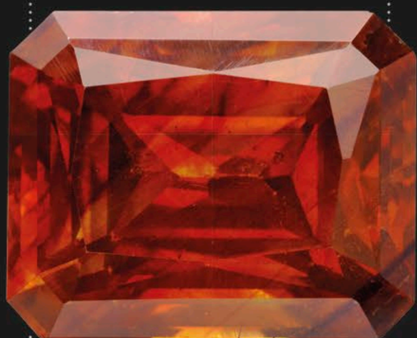
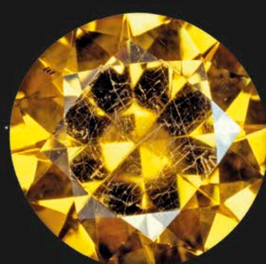
Fire and brilliance

The term "fire" is used to describe the flashes of light that make a gemstone sparkle when it is moved. As with a prism, when white light enters a gem its component colours are dispersed: the greater the dispersion of white light, the greater the fire. The refractive index (RI, see p.23) is a measure of dispersion. Diamonds have high dispersion and are valued for their brilliance; however, gems with low RI can be valued for other reasons.

Diamond RI 2.42
Cut diamonds have a very high RI and great dispersion, giving the gems their distinctive brilliance and shine.



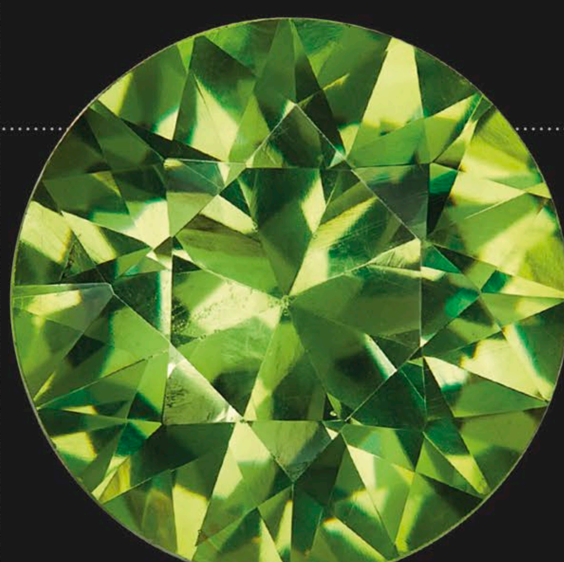
Sphalerite RI 2.36–2.37
Sphalerite is extremely difficult to facet, but cut stones exhibit a high RI and good fire.



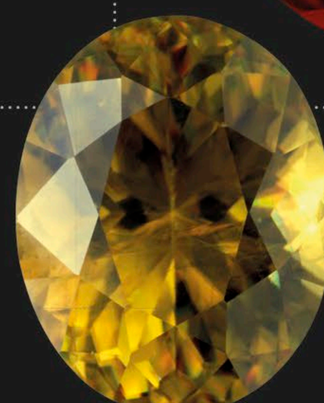
Scheelite RI 1.92–1.93
Faceted scheelite is rare. However, it exhibits good dispersion of light when cut.

Cassiterite RI 2.00–2.10
Cassiterites are dichroic, displaying different colours when moved.

Zircon RI 1.81–2.02
Zircon's high RI and excellent dispersion come close to diamond in terms of fire and brilliance.



Demantoid garnet RI 1.85–1.89
This variety is the most highly sought-after type of andradite garnet; it has greater colour dispersion than diamond.



Sphene RI 1.84–2.11
Transparent sphene crystals, where they occur, have good fire and brilliance.